SYS 304: Trade Studies, Risk, and Decision Analysis

Decision Making Under Risk and Uncertainty

Homework 2 Solution

1. A typical problem involving five alternatives and four states of nature together with associated probabilities is shown in the table below. The numbers in the table can be thought of as returns or profits in thousands of dollars of net PW, such that a negative number is a cost.

Alternatives	State of nature / (probability of state occurring)			
	S_1 (0.5)	S_2 (0.1)	S_3 (0.1)	S_4 (0.3)
A_1	3	-1	1	1
A_2	4	0	-4	6
A_3	5	-2	0	2
A_4	2	-2	0	0
A_5	5	-4	-1	0

a. Check for dominance and take appropriate action. From this point forward include in your analysis only the remaining alternatives after dropping the dominated ones.

Use the dominance criterion and perform mutual comparison for all 10 pairs of alternatives (A_i, A_j). It can be observed that:

- Alternative A_1 dominates alternative A_4
- Alternative A_3 dominates alternative A_4 – The comparison is not needed, since A_4 was already eliminated.
- Alternative A_3 dominates alternative A_5

Thus, alternatives A_4 and A_5 can be both eliminated from further consideration.

Alternatives	State of nature / (probability of state occurring)			
	S_1 (0.5)	S_2 (0.1)	S_3 (0.1)	S_4 (0.3)
A_1	3	-1	1	1
A_2	4	0	-4	6
A_3	5	-2	0	2

b. Determine which would be the best alternative if the decision maker has each of the following aspiration levels: (1) possible result of at least 5; (2) possible negative result (loss) no worse than -1.

Alternatives	State of nature / (probability of state occurring)			
	S_1 (0.5)	S_2 (0.1)	S_3 (0.1)	S_4 (0.3)
A_1	3	-1	1	1
A_2	4	0	-4	6
A_3	5	-2	0	2

(1) Alternative A_2 and A_3 have possible results of 5 or greater.
If further selection is needed, select A_3 since probability of future to occur is greater.
(2) Alternative A_1 has a possible negative result no worse than -1.

c. Determine which would be the best alternative using the most probable future criterion.

Alternatives	State of nature / (probability of state occurring)			
	S_1 (0.5)	S_2 (0.1)	S_3 (0.1)	S_4 (0.3)
A_1	3	-1	1	1
A_2	4	0	-4	6
A_3	5	-2	0	2

Most probable future is S_1 having a probability of 0.5.
Alternative A_3 has the highest payoff for future S_1 and is the one selected.

d. Show which alternative is best by the expected value criterion.

Alternatives	State of nature / (probability of state occurring)			
	S_1 (0.5)	S_2 (0.1)	S_3 (0.1)	S_4 (0.3)
A_1	3	-1	1	1
A_2	4	0	-4	6
A_3	5	-2	0	2

$E[A_1] = (3)(0.5) + (-1)(0.1) + (1)(0.1) + (1)(0.3) = 1.8$
$E[A_2] = (4)(0.5) + (0)(0.1) + (-4)(0.1) + (6)(0.3) = 3.4$
$E[A_3] = (5)(0.5) + (-2)(0.1) + (0)(0.1) + (2)(0.3) = 2.9$

Alternative A_2 is selected.

e. Determine which alternative is best using the Laplace rule.

Alternatives	State of nature / (probability of state occurring)			
	S_1 (0.5)	S_2 (0.1)	S_3 (0.1)	S_4 (0.3)
A_1	3	-1	1	1
A_2	4	0	-4	6
A_3	5	-2	0	2

$E[A_1]$ = (3)(0.25) + (-1)(0.25) + (1)(0.25) + (1)(0.25) = 1.0
$E[A_2]$ = (4)(0.25) + (0)(0.25) + (-4)(0.25) + (6)(0.25) = 1.5
$E[A_3]$ = (5)(0.25) + (-2)(0.25) + (0)(0.25) + (2)(0.25) = 1.25

Alternative A_2 is selected.

f. Determine which alternative maximizes the minimum possible payoff.

Alternatives	State of nature / (probability of state occurring)				Maximin Criterion
	S_1 (0.5)	S_2 (0.1)	S_3 (0.1)	S_4 (0.3)	
A_1	3	-1	1	1	-1
A_2	4	0	-4	6	-4
A_3	5	-2	0	2	-2

Alternative A_1 is selected.

g. Determine which alternative maximizes the maximum possible payoff.

Alternatives	State of nature / (probability of state occurring)				Maximax Criterion
	S_1 (0.5)	S_2 (0.1)	S_3 (0.1)	S_4 (0.3)	
A_1	3	-1	1	1	3
A_2	4	0	-4	6	6
A_3	5	-2	0	2	5

Alternative A_2 is selected.

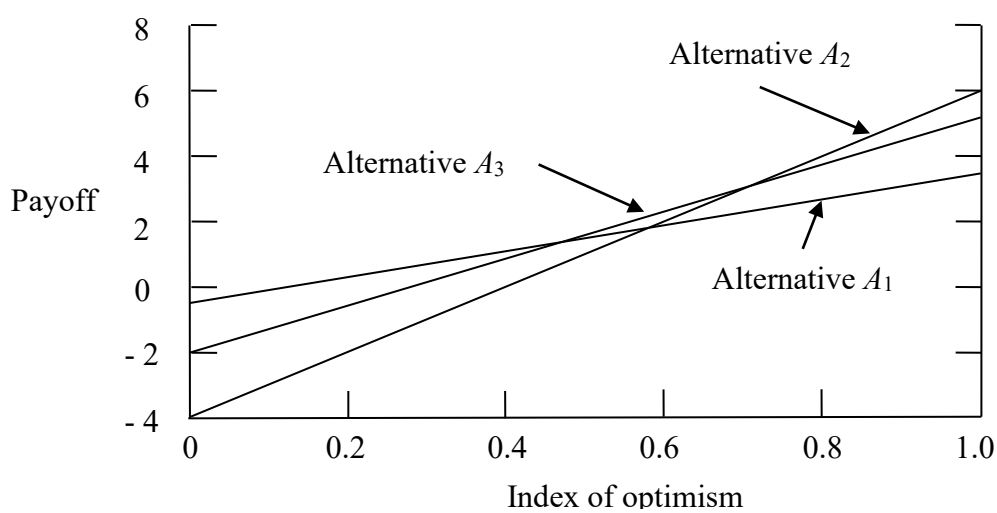
h. Show which alternative would be chosen on the basis of minimizing the maximum regret.

Alternatives	State of nature / (probability of state occurring)				Minimax Regret Criterion
	S_1 (0.5)	S_2 (0.1)	S_3 (0.1)	S_4 (0.3)	
A_1	3 5 – 3 = 2	-1 0 + 1 = 1	1 1 – 1 = 0	1 6 – 1 = 5	5
A_2	4 5 – 4 = 1	0 0 – 0 = 0	-4 1 + 4 = 5	6 6 – 6 = 0	5
A_3	5 5 – 5 = 0	-2 0 + 2 = 2	0 1 – 0 = 1	2 6 – 2 = 4	4

Alternative A_3 is selected.

i. Calculate which alternative would be best using the Hurwicz rule, for an index of optimism of 0.75. Also, graph the calculated payoff for each alternative over the entire range of the index of optimism.

Alternatives	State of nature / (probability of state occurring)				Maximax Criterion
	S_1 (0.5)	S_2 (0.1)	S_3 (0.1)	S_4 (0.3)	
A_1	3	-1	1	1	3
A_2	4	0	-4	6	6
A_3	5	-2	0	2	5

$E[A_1] = (0.75)(3) + (0.25)(-1) = 2.0$
$E[A_2] = (0.75)(6) + (0.25)(-4) = 3.5$
$E[A_3] = (0.75)(5) + (0.25)(-2) = 3.25$

Alternative A_2 is selected.